

Major Project 3

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ENG102

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Chapter One

Introduction

Machine learning algorithms have become a crucial part in the data science community. As the amount of data continues to increase as time goes by, data scientists feel the importance of computation skills to extract summaries, make predictions, and organize big data. Machine learning types are generally grouped into supervised learning algorithms, unsupervised learning algorithms, and reinforcement learning. Supervised learning algorithms include linear regression, logistic regression, decision trees, gradient boosting, and neural networks. Unsupervised learning algorithms consists of clustering, principal component analysis, T-distributed stochastic neighbor embedding(T-SNE), and Independent Component Analysis. Reinforcement learning involves Markov decision processes(MDPs), Monte Carlo Tree Search, and Q-Learning.

The difference between these three machine learning algorithms are supervised machine learning algorithms utilize labeled data (data that has outputs) to make predictions, unsupervised learning algorithms identify hidden structures in unlabeled data (input only and no correct outputs/labels attached, for example, image data), while reinforcement learning concentrates on make decisions based on rewards and penalties.

Research Question

The research question that is interested to be delve deeper into in this paper is “what role do machine learning algorithms play in the data science discourse community?”. This is especially important since data science integrates statistics and computing in solving tasks.

The Data Science Discourse Community

Discourse Community

Discourse communities are groups that share common goals and fulfill certain characteristics mentioned by both the weekly reading and video. They can be formed by various groups, such as general education classes, college majors, divisions in companies and more. Melzer (2020), in his essay, explains that cohorts have to fulfill six features to be considered as discourse communities, which they should have common public goals that are agreed among their members; utilize forms of communications; use these communications to share information and feedback; involve one or more genres to help reach the goal of their communities; acquire some specific lexis, jargon or terminology; and have a threshold level of expert members (p. 102).

The Goal of the Data Science Discourse Community

The common goal of the data science community is to extract insights from data to solve real world problems. Melzer (2020), in his essay, explains that cohorts have to fulfill six features to be considered as discourse communities, which they should have common public goals that are agreed among their members; utilize forms of communications; use these communications to share information and feedback; involve one or more genres to help reach the goal of their communities; acquire some specific lexis, jargon or terminology; and have a threshold level of expert members (p. 102).

The Mechanisms of Communication of the Data Science Discourse Community

Data science community members use various mediums to communicate with one another, for example, through GitHub, Discord channels, dashboards, blogs, and online forums. GitHub is specifically used to showcase work that has been done by each data scientist. It can be used to help people acquire jobs since employers can view actual projects that they have done. Discord channels are typically used among members to create group 3 chats or to keep in touch with one another for conversations related to work or personal matters such as making friends. Online forums and blogs are more like public places to ask questions and answer people's inquiries, which include topics such as asking for feedback when one faces an error in their code, asking suggestions about which company they should apply for, or improvements that can be made in their projects. Dashboards are usually the final products of data scientists' work that are summarized in visual representations.

Genres or Unique Texts to The Data Science Discourse Community

Genres, which are unique texts to each discourse community, are created to meet the needs of different discourse communities. Among the above mechanisms of communication, the most obvious ones that can be considered genres are data dashboards and GitHub. This is because, referring to the definition of genres in the context of discourse community as texts that are unique to the discourse community and its members, dashboards and Github fulfill the criteria. Github most of the time stores codes, which generally only communities that do coding use this communication channel, while dashboards are very identical with data scientists since they are the summary of data.

Specialized Language or Lexis in the Data Science Community

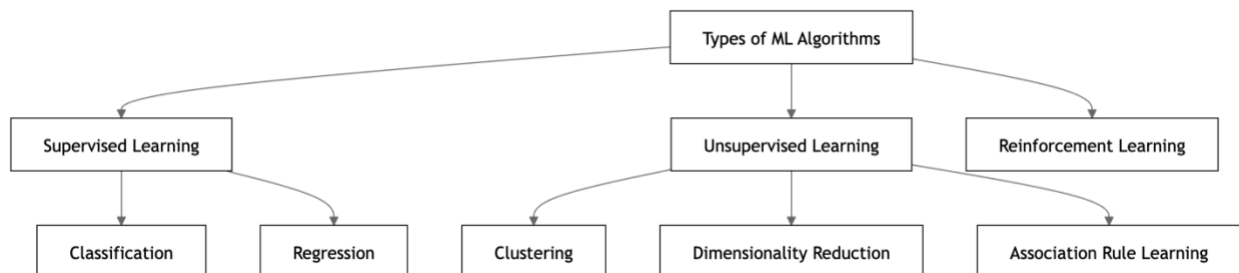
The kind of specialized language or lexis that this discourse community members use in their conversation and genres consists of various technical terms. It involves data wrangling, data cleaning, data mining, feature engineering, exploratory data analysis, regression, machine learning, neural network, XGBoost(Extreme Gradient Boosting), training data, validation data, cross validation, accuracy, confusion matrix, evaluation tables. People 4 that have been involved in this community and actively work on actual data science projects usually know these terms. These terms are used to communicate with fellow data scientists about certain steps that they want to discuss further.

New Comers and Experts in the Data Science Community

The old-timers in the field of data science are senior data scientists, professors, industry leaders. On the other hand, newcomers are students, people who have just switched their careers to data science, and entry-level data scientists. Novice data scientists can obtain knowledge of data science through various sources, such as online tutorials, data science-related websites, blogs, forums, and online courses

Chapter Two

Machine Learning Algorithms in Data Science



Supervised Learning

Supervised learning is a type of machine learning algorithms in which models are trained using labeled data, which conveys that each input in a dataset has a corresponding output. The goal of supervised learning is to make accurate predictions on unseen data. Some examples of supervised learning problems are regression, where the target variable is numerical, such as predicting house prices, and classification, where the output is categorical (such as identifying whether a customer will reorder an item or not). The examples of supervised learning algorithms are linear regression, logistic regression, decision trees, random forest, support vector machine (SVM), K-Nearest Neighbors(KNN), neural networks, and gradient boosting.

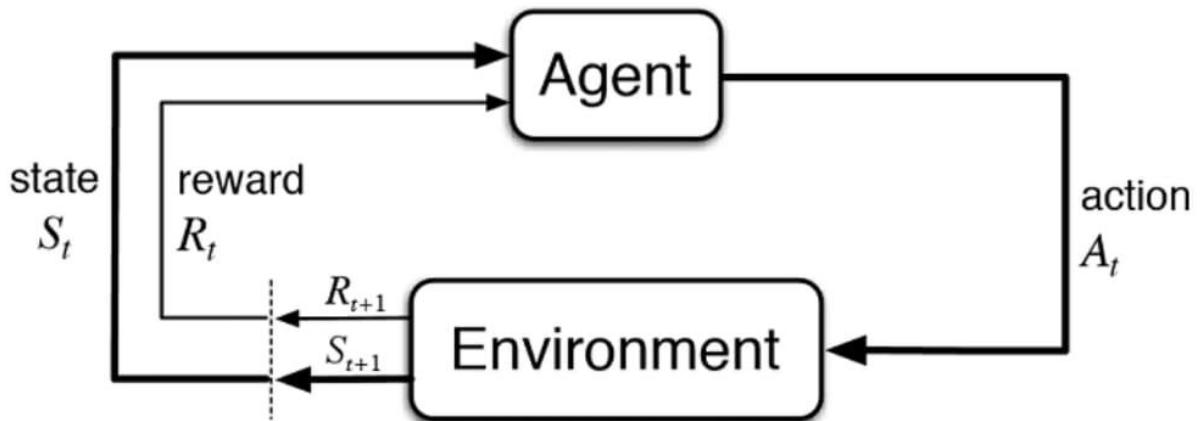
Unsupervised Learning

Unsupervised learning is a category of machine learning algorithms that works with unlabeled data, which outputs or target variables are not given in the dataset. It instead tries to find patterns or relationships with the given dataset. Common tasks in unsupervised learning consist of clustering in which data with similar attributes are grouped together, for purposes such as to group customers into categories to better learn their buying habits and dimensionality reduction, where the number of features is reduced for interpretability purpose and to reduce over fitting. For instance, K-Means Clustering, hierarchical clustering, principal component analysis (PCA), t-distributed stochastic neighbor embedding(t-SNE), and association rule learning.

Reinforcement Learning

Reinforcement learning is a type of machine learning algorithms where an agent (the learner and the decision maker) studies by using the concepts of rewards and penalties based on actions taken by it. As it practices, the agent learns which actions produce the highest total

reward and to avoid actions that cause penalties. The example of reinforcement learning includes artificial intelligence in the chess game.



Evaluation Metrics

Evaluation metrics are used to assess models in data science. There are a lot of evaluation metrics that data scientists can use to measure the performance of algorithms and models that they use in their work. Some examples of widely used evaluation metrics are confusion matrix, mean absolute error (MAE), mean squared error (MSE), receiving operating characteristics (ROC) curve, area under the curve (AUC). Below are some examples of how evaluation metrics that can be utilized in data science.

Accuracy. Accuracy measures the overall performance of the model based on the total number of true positives, true negatives, false positives and false negatives. It calculates the number of correct predictions, which are true positives and true negatives, and divides them by the total number of predictions. Accuracy can be used when the dataset is balanced, which the total counts of all groups are similar.

$$\frac{TRUE POSITIVES + TRUE NEGATIVES}{TRUE POSITIVES + TRUE NEGATIVES + FALSE POSITIVES + FALSE NEGATIVES}$$

Precision. Precision communicates how accurate the positive predictions are. It is calculated by dividing the number of true positives by true positives and false positives. High precision value means the model makes a high number of correct predictions.

$$\frac{TRUE POSITIVES}{TRUE POSITIVES + FALSE POSITIVES}$$

F1 Score. This is a combined metric, that is compatible for unbalanced classes, as it considers both false positives and negatives. It ranges from 0 to 1, with a higher score indicating better model quality. F1 Score demonstrates how well a model performs in identifying positive cases while minimizing false positives and false negatives.

$$\frac{2 * TRUE POSITIVES}{2 * TRUE POSITIVES + FALSE POSITIVES + FALSE NEGATIVES}$$

Common Evaluation Metrics.

Metric	Category	Description	Formula / Notes
Accuracy	Classification	Proportion of correct predictions	$(TP + TN) / (TP + TN + FP + FN)$
Precision	Classification	Proportion of positive predictions that are correct	$TP / (TP + FP)$
Recall (Sensitivity)	Classification	Proportion of actual positives	$TP / (TP + FN)$

		correctly identified	
F1 Score	Classification	Harmonic mean of precision and recall	$2 * (\text{Precision} * \text{Recall}) /$ $(\text{Precision} + \text{Recall})$
Specificity	Classification	Proportion of actual negatives correctly identified	$\text{TN} / (\text{TN} + \text{FP})$
ROC-AUC	Classification	Area under ROC curve	Measures separability of different classes(positive or negative)
Confusion Matrix	Classification	Table of TP, TN, FP, FN	TP, TN, FP, FN
Mean Absolute Error (MAE)	Regression	Average absolute difference between predicted and actual values	$1/n * \sum y - \hat{y} $
Mean Squared Error (MSE)	Regression	Average squared difference	$1/n * \sum (y - \hat{y})^2$

Root Mean Squared Error (RMSE)	Regression	Square root of MSE	$\sqrt{\text{MSE}}$
R-squared (R^2)	Regression	Proportion of variance explained	$R^2 = 1 - [\Sigma(y_i - \hat{y}_i)^2 / \Sigma(y_i - \bar{y})^2]$
Adjusted R^2	Regression	Adjusted for number of predictors	Penalizes multivariate variables
Mean Absolute Percentage Error (MAPE)	Regression	Average percentage error	$100/n * \Sigma (y - \hat{y})/y $

Applications of Data Science

Financial or Businesses

Companies can improve efficiency, profitability, and effectivity through implementing data science techniques. By applying statistical and computing methods, raw data can be transformed into meaningful insights for policy makers to create better decisions. According to Ullah et al. (2025), based on firm historical data, data scientists can develop data-driven models that estimate customer/user actions and find patterns (p. 289).

One of the examples of data science and machine learning applications can be seen in companies analyzing customer data to understand purchasing behavior and customizing

marketing strategies using firm data that has been grouped into clusters, which allows business to design campaigns that enhance companies' performance.

Smart Cities or Urban

In smart cities, machine learning models analyze data from street cameras, Global Positioning System (GPS), and traffic sensors to optimize traffic flow. Street conditions can be updated based on real-time conditions to maximize travel routes and fuel consumptions.

Additionally, public transportation systems also receive a significant benefit from data science, as predictive models are able to estimate passenger demand at different locations, which allows people to decide which public transportation they should take. Moreover, machine learning can analyze energy consumption in smart cities to reduce global warming. Ullah et al. (2025) believes that cities are being transformed into safer, greener, and more efficient communities through the implementations of sensors, cameras, and data analytics (p. 289).

The Rise of Machine Learning for Big Data Processing

Data scientists nowadays incorporate machine learning into their workflows more than in the past due to its ability to handle massive datasets more quickly than traditional methods. As data becomes larger and more complex, older statistical approaches alone are often insufficient to process it in a timely manner. In contrast, machine learning models can handle large volumes of data rapidly and improve accuracy. According to Ullah et al. (2025), machine learning techniques have been applied across various industries, such as finance, agriculture, and logistics, to enhance company systems.

Machine Learning and Advancement in Technology

As new models and techniques in data science continue to emerge, complex problems are becoming easier and faster to solve. Luo et al. (2025) argue that innovations such as Graph

Neural Networks (GNNs) and Generative Adversarial Networks (GANs) demonstrate how complex tasks involving logistics optimization and network analysis can now be solved using data science tools. These advancements in the data science discourse community illustrate that it continues to evolve and grow alongside emerging technologies developed for data scientists.

Chapter Three

Primary Data Overview

The primary data contains around 600 observational units upon entering the data cleaning step. This paper explores the role of machine learning within the data science community. It aims to understand how widely machine learning is used, which algorithms are most common, and what topics are frequently discussed among community members. The study uses data from the Stack Exchange forum, which was cleaned and analyzed using R Studio. Through this process, the research targets to explain insights obtained from the primary data that was selected to enrich this community.

The research project is organized into a timeline across several weeks. One week to collect data and another one week to further analyze the data. The research question that contributes as the foundation of this chapter is the role of machine learning in the data science community. The author selected text analysis as the primary research method for this chapter. This method was chosen because it enables the analysis of a large number of individuals through texts that exist in this community, which allows broader observations. It also demonstrates one of the data science applications in processing data.

Methodology

First, the author collected the raw data from the Stack Exchange community forum about the data science community. The data was then cleaned, which involves handling missing values.

Subsequently, the data was further explored through graphical and numerical summaries.

Afterwards, the visualizations were interpreted for the general audience.

Anticipated Challenges

The challenges will be the time taken to realize all of the ideas of the author of this paper into a tangible action and the relevance of my data source since people started to abandon online community forums as a consequence of the development of Artificial Intelligence. People might not see online community forums as the first thing that comes to mind whenever they have a question.

Top 5 by Score

Question.Id	Title	Score	View.Count	Answer.Count	Owner.Display.Name	Creation.Date	Link
1	19048 What is the difference between test set and validation...	570	675608	12	xiaohan2012	2011-11-28	https://stats.stackexchange.com/questions/19048/...
2	133656 How to understand the drawbacks of K-means	448	180054	5	KevinKim	2015-01-16	https://stats.stackexchange.com/questions/133656/...
3	99171 Why is Euclidean distance not a good metric in high d...	366	172636	8	teaLeef	2014-05-18	https://stats.stackexchange.com/questions/99171/...
4	222179 How to know that your machine learning problem is h...	301	37255	3	Tim	2016-07-05	https://stats.stackexchange.com/questions/222179/...
5	7207 ROC vs precision-and-recall curves	273	139185	7	Amelio Vazquez-Reina	2011-02-14	https://stats.stackexchange.com/questions/7207/ro...

Top 5 by Date

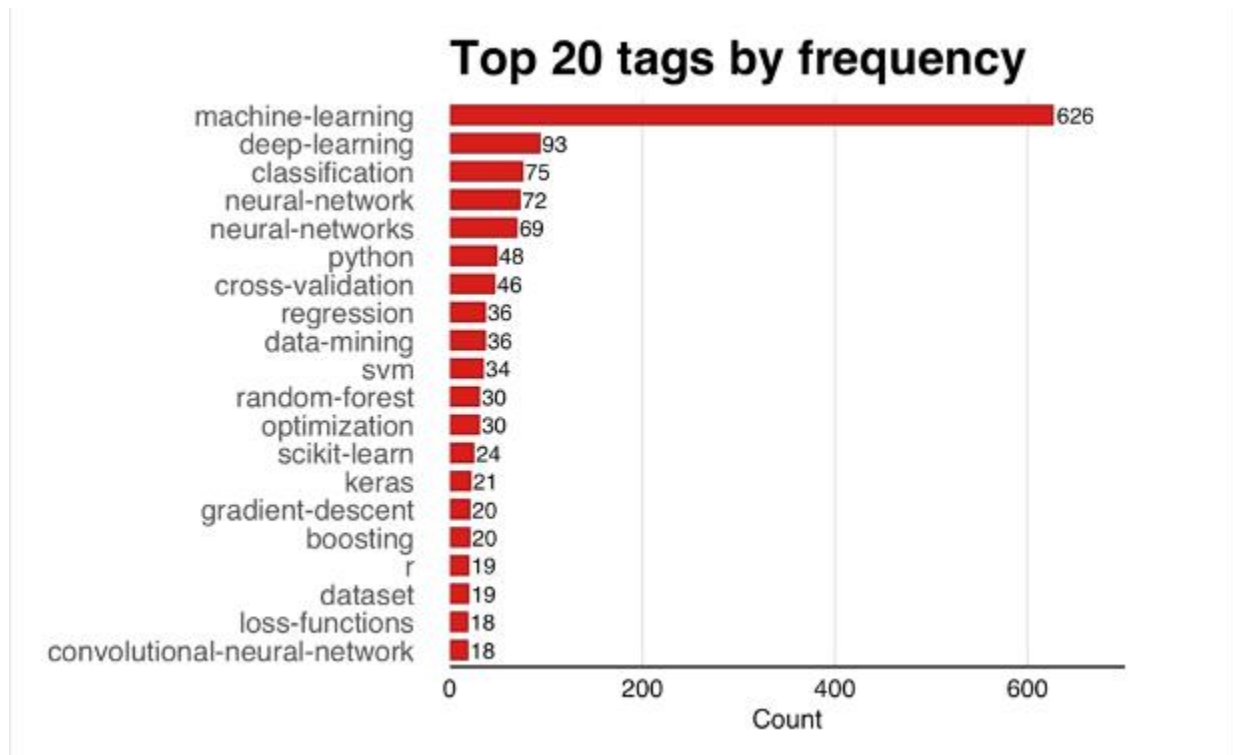
Question.Id	Title	Score	View.Count	Answer.Count	Owner.Display.Name	Creation.Date	Link
1	136572 Use of training data that has been labeled by the AI m...	12	1617	1	raner	2025-11-03	https://datascience.stackexchange.com/questions/1...
2	131488 In training a neural network, why don't we take the d...	12	2041	4	Leo Juhlin	2025-03-03	https://datascience.stackexchange.com/questions/1...
3	126337 Why does data science see class imbalance as a probl...	23	1313	2	Dave	2024-01-09	https://datascience.stackexchange.com/questions/1...
4	120764 How does an LLM "parameter" relate to a ...	18	19966	1	slim_wizard	2023-04-06	https://datascience.stackexchange.com/questions/1...
5	590861 What is the difference between something being &qu...	33	5745	5	stochasticmrf	2022-10-02	https://stats.stackexchange.com/questions/590861/...

Findings and Interpretation

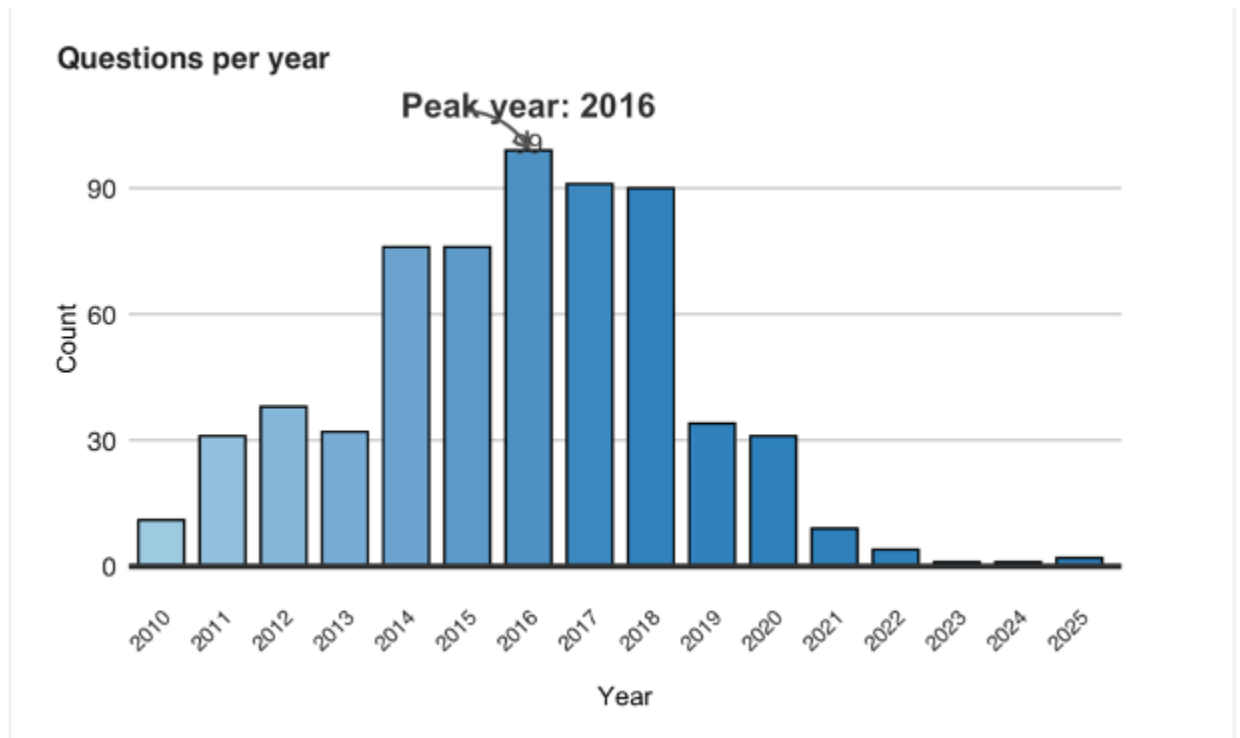
The texts differ in their focus across distinct stages of the data science workflow. Questions in the top 5 by score and the top 5 by date discuss different phases, including data cleaning, data wrangling, exploratory data analysis, feature engineering, predictive modeling, and data visualization. This variation highlights that members in this discourse community tend to ask questions on specific steps rather than the whole workflow.

The questions include: “What is the difference between test set and validation set?”, “How to understand the drawbacks of K-means”, “How to know that your machine learning problem is hopeless?”, “ROC vs precision-and-recall curves”, “Use of training data that has been labeled by the AI model itself”, “In training a neural network, why don’t we take the derivative with respect to the step size in gradient descent?”, “Why does data science see class imbalance as a problem for supervised learning when statistics does not?”, “How does an LLM ‘parameter’ relate to a ‘weight’ in a neural network?”, “What is the difference between something being ‘true’ and ‘true with probability 1’?”

The findings reveal that conceptual understanding still becomes the main question even until today. It is illustrated by how most of the questions that were asked by members in this discourse community concentrate on the “why” instead of just “what”. Model evaluation and algorithms are the central questions in this community. Additionally, the discussions are mostly about machine learning both based on the highest scores and recent dates, which portrays how widely used machine learning is in this discourse community. The texts also depict that members put a heavy emphasis on interpretation and real world implementations on top of algorithms and model performance metrics, such as when handling issues like class imbalance. Overall, this community is centered towards both theories and applications.



The graph shows the top 20 tags by frequency. The x-axis represents the total count, while the y-axis depicts the tags that consist of various topics in the data science discourse community. Some questions might contain more than 1 tag. The most dominant tag is machine-learning, with the total count of 626 tags, indicating its popularity in this discourse community. This is followed by deep-learning(93) and classification(75). Middle tags like python, cross-validation, regression, and data-mining suggest model evaluations and algorithms are also crucial themes. Lower tags such as keras, gradient-descent, boosting, and convolutional-neural-network demonstrate that more advanced topics that are discussed less often. Overall, the graph reflects the primary focus of this community is on machine learning fundamentals, followed by techniques and tools that are typically used by data scientists.



The chart shows the number of questions asked per year from 2010 to 2025. The x-axis depicts the years (2010-2025), whereas the y-axis demonstrates the total count of questions asked in certain years. The number of questions increased steadily from 2010, and reached its peak in 2016 (around 100 questions in total). It then stayed relatively high until 2018, before it experienced a significant drop afterward. The most non-active period of this discourse community appear to be from 2022-2025. This is mainly because during those years, artificial intelligence had been developed, which made some members prefer to ask their questions in artificial intelligence chat bots rather than in online forums.

Connections with the Secondary Research Findings

The trends above are further discussed by Sarker (2021), who explains that machine learning algorithms are not only widely used but also applied in actual scenarios such as sentiment analysis and predictive modeling. The primary data supports this claim, as many of the highly scored and recent questions focus on practical implementation topics, including

evaluation metrics like ROC curves, class imbalance, and neural network. This demonstrates that members of the discourse community actively engage with both theoretical and applied aspects of machine learning.

Alshuaibi et al. (2025) explain that machine learning is increasingly applied in specialized fields such as cybersecurity, which demonstrates its role across different domains. This supports the observation from the primary data findings that discussions are not limited to basic concepts but also to diverse implementations of machine learning.

Luo et al. (2025) also argue that technologies such as Graph Neural Networks (GNNs) and Generative Adversarial Networks (GANs) enable more complex problem-solving. This is consistent with how advanced topics are frequently less discussed in this discourse community, which suggests that while foundational concepts are often brought up, the discourse community is also gradually incorporating newer techniques.

Correlations of Findings to the Claim or Thesis

The results from the primary data indeed support what have been concluded in the secondary research analysis and theories have been used in this paper. The high total count of machine learning tags, along with deep learning, classification, and neural networks represents that machine learning is not only commonly used in this community but also a primary discussion among members in this discourse community.

Furthermore, the variation in topics across different stages of the data science workflow, including data cleaning, data wrangling, exploratory data analysis, feature engineering, predictive modeling, and visualization shows that machine learning is integrated throughout the entire process rather than being limited to a single stage. Overall, machine learning has become a main component that influences discussions and approaches in this community. Hence, this

clearly resonates with the thesis statement in earlier chapters that machine learning plays an extremely critical role in the data science discourse community.

Annotated Bibliography

Ullah, I., Adhikari, D., Su, X., Palmieri, F., Wu, C., & Choi, C. (2025). Integration of data science with the intelligent IoT (IIoT): current challenges and future perspectives. *Digital Communications and Networks*, 11(2), 280-298.

The article explains how data science is implemented across various domains, such as finance, retail industry, smart cities, logistics management, warehouses, food services, agriculture to enhance their systems. It starts off by defining the definition of data science and providing explanations of terms used in this paper, and it goes on to examine diverse methods that can be used in different cases.

This journal was published in 2025, making it highly relevant to the modern era. Not only is this paper recent, but it also includes findings from over 140 studies. It breaks down technologies that are unpopular into different sections, so people are able to digest the information better. The limitations of this article being data for the findings were collected through a survey, which makes the methods that were stated in this paper to be unproven in an experiment and the discussion of each topic is not too detailed since this journal covers many concepts.

This article helps me comprehend the data science discourse community, as it shows distinct workplaces that members in this community work in, and the evolution of some domains that integrate data science into their applications to maximize their potential. It also provides multiple approaches that can be used in handling data science problems.

Sarker, I. H. (2021). Machine learning: Algorithms, real-world applications and research directions. *SN computer science*, 2(3), 160.

The article provides a thorough analysis on various machine learning algorithms in data science including supervised, unsupervised, semi-supervised, and reinforcement learning. It also attaches diagrams to demonstrate complex processes or methods. Applications of methods stated in the journal are also explained with actual techniques that are available to use. This source is relevant to my research, as it ties machine learning concepts to practical applications and assesses types of data that each model can take. The explanations in each session are also extensive in this paper.

For example, this paper states real company names that utilize similar algorithms in doing projects stated in this article. However, since the paper discusses contents in depth, there might be plenty of terms that can be difficult to understand for readers without a computing background. This journal assists me in learning advanced algorithms that can be beneficial if used correctly and have been used in this discourse community. For example, the use of natural language processing in sentiment analysis to identify public mood and views on reviews, social media, forums, and news.

Alshuaibi, A., Almaayah, M., & Ali, A. (2025). Machine learning for cybersecurity issues: A systematic review. *Secur Challenges*, 1(2).

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Theodorakopoulos, L., Karras, A., & Krimpas, G. A. (2025). Optimizing apache spark MLlib: Predictive performance of large-scale models for big data analytics. *Algorithms*, 18(2), 74.

This paper goes over the use of Apache Spark in optimizing the predictive performance of large data for big data analysis. It investigates the performance of machine learning algorithms, for instance, K-Means and Random Forest Regression in Apache Spark. Apache Spark is an open-source framework that is designed for big data. It includes evaluation metrics and diagrams to show the strength of Apache Spark in analyzing large-volume data. The article is also very recent, making claims in this paper to be encouraging.

However, as this article only uses several algorithms in evaluating the performance of Apache Spark, other algorithms might perform differently in the open-source framework. This journal helps me to understand the data science discourse community better as it portrays ways to evaluate models using different evaluation metrics and diagrams as well as integrating statistical threshold values such as

R^2 (coefficient of determination) to assess the total time and memory usage for each model validated in Apache Spark.

Luo, H., Wei, J., Zhao, S., Liang, A., Xu, Z., & Jiang, R. (2025). Intelligent logistics management robot path planning algorithm integrating transformer and gcnn network. arXiv preprint arXiv:2501.02749.

This paper discusses modern technologies in data science that are not used as often as other techniques in data science, such as regression, classification, decision trees, namely Graph Neural Networks(GNN) and Generative Adversarial Network(GAN). This paper discusses the application of machine learning in path management algorithms. It includes path length, time efficiency, and energy consumption variables to construct models. This is extremely relevant for the members in this discourse community that intend to implement their data science skills in the logistics and distribution field.

Additionally, this paper involves mathematical formulas to assess models that were used in this paper. However, this paper does not really explore how hardware equipment, such as battery capacity and equipment power affect the implementation of methods proposed in the paper. This paper aids me in understanding the data science discourse community by providing an application example of data science in logistics management to elevate efficiency and productivity.

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